Automation and Control Laboratory

Prof. Alessio La Bella
DEIB – Politecnico di Milano

AY 2025/26 – II semester

General info

Teacher

Alessio La Bella
DEIB – Building 20
alessio.labella@polimi.it

Tutor

Mar Puchades Ibáñez
mar.puchades@polimi.it

Office hours

By appointment (ask via e-mail)

Course website

WeBeep Portal (https://webeep.polimi.it/)

Course info

P.za L. Da Vinci prima di Via Ampere
Via Bassini prima di Via Ponzio
Via Bonardi prima di Via Ponzio
Via Edoardo Bonardi
Via Edoardo Bonardi
P.za L. Da Vinci Fronte 9 prima di Via Spinoza
CT1
10
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Edificio 7
Carlo Erba
Via Giuseppe Ponzio
Piazza Leonardo da Vinci
Piazza Leonardo da Vinci
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Piazza Leonardo da Vinci
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Viale Dottor Architetto Luigi Birarda
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9A
Via Giuseppe Ponzio
Via Giovanni Celoria
Via Giovanni Celoria
Squash

Course info

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Course objective

Challenge the students in applying the theoretical skills acquired during the study program on a real apparatus

Topics

Span the design/development/deployment of automated systems:

- system modeling and analysis
- controller synthesis and analysis of the closed-loop system
- implementation on a real apparatus
- validation of the model/controller performance

Format

Numerical simulations and real experiments via a *team project work*

Prerequisites

Students are expected to have previous knowledge about

- mechanical/hydraulic systems modeling and dynamics
- fundamentals of control theory
- electrical actuators and sensors
- system identification
- advanced and multivariable control techniques

Laboratory material

In the laboratory there are different set-ups, each one simulating (on a small scale) a real system to be controlled

Available control problems

- control of unstable systems
- control of marginally stable systems
- control of underdamped systems

Organization info

Laboratory availability

Monday: 2:30 pm → 7:00 pm (5 hours)

Tuesday: 2:30 pm → 7:00 pm (5 hours)

Friday: 8:30 pm → 1:00 pm (5 hours)

Course organization

- students are asked to form teams of 4 people each
- each group express a preference regarding the set-up it wants to work on and a preference regarding the day
- we will then (to our best) assign set-ups and days to groups
- each team will work autonomously on the assigned set-up (with the support of a teacher/tutor)
- all team members are required to be in the lab

Expected steps

- preliminaries
  - model of the set-up at hand
  - identification of those parameters that are difficult to measure (friction coefficient, spring stiffness, etc.)
  - model validation via time and frequency responses
  - open-loop analysis (stability, observability, controllability, operating range, performance)
- list of self-given specifications for the close-loop system
- for each control task
  - design based on a linearized/nonlinear model
  - test of the controller on the model via simulation
  - controller validation via time and frequency responses
- critical analysis/comparison of all developed control strategies

Control techniques

- Frequency-Based (FB) techniques
  - compute the transfer function $G(s)$ of the linearized system
  - design the $R(s)$ such that $L(s)$ has desired properties
  - design via Bode diagrams (or root locus or Nyquist)
- State-Space (SS) techniques
  - compute the state-space representation of the linearized system
  - design state estimators (Luenberger observer & Kalman filter)
  - design Pole-Placement & Linear-Quadratic controllers
- Advance-Control (AC) techniques, for example:
  - Model Predictive Control (MPC)
  - Sliding Mode
  - Feedback Linearization
  - Adaptive/Robust Control

Evaluation info

Time validation

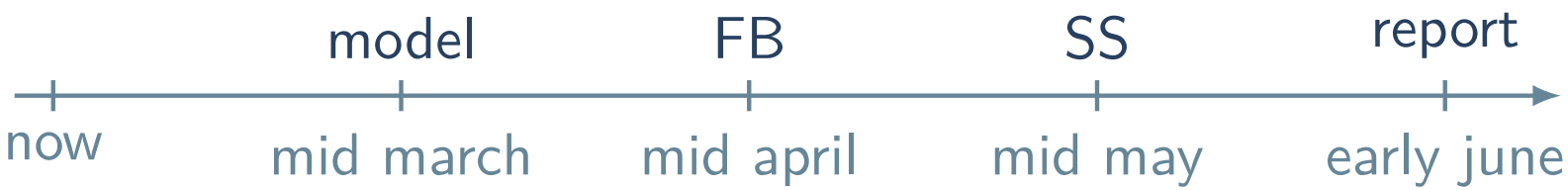
- input impulse/step to $u(t)$ (or $y^\circ(t)$ in close loop)
- record output response $y(t)$ (and $u(t)$ in close loop)
- compare with model impulse/step response
  in terms of gain, settling time, ...

Frequency validation

- input sine waves to $u(t)$ (or $y^\circ(t)$ in close loop)
- record output (sine) responses $y(t)$ (and $u(t)$ in close loop)
- compare with model frequency response on a Bode diagram

Evaluation info

Expected timeline



Suggestions

- plan activities for the next lab session
- have a backup plan

Evaluation info

Team duties

- final report (max 40 pages, 🇮🇹 / 🇬🇧) describing all the work done by the team on the assigned set-up
- final presentation (20 min + 20 min Q&A, 🇬🇧) to the whole class describing the work done by the team

Evaluation criteria

- participation to the team activity during laboratory hours
- content, clarity, and technical soundness of the final report
- content and clarity of the final presentation
- overall approach to the lab experience

Teacher/Tutor feedback

Mid-term review

Halfway through the semester there will be a 30 minutes private review meeting with each group

Format

You will be asked to discuss (free format, /) the result achieved so far and a planned schedule for the future work

Outcome

The meeting is not graded: it is only meant to give you feedback about your progress

Summary

Key dates

Feb 27 @ 2p.m: deadline for the "preference form" submission

Mid. Apr: review meetings (a schedule will follow)

Jun 7 or 14: deadline for the final report submission

Jul 1: final presentations (during the exam period)

Note

Attendance of all students to all presentations is recommended

Preference form

Each team has to provide (by email) a preference form containing:

- team nickname
- ID (Codice Persona), name, and email of each component
- a ranking of all available set-ups/days, ordered by preference

The preference form can be downloaded from the WeBeep portal

Available set-ups

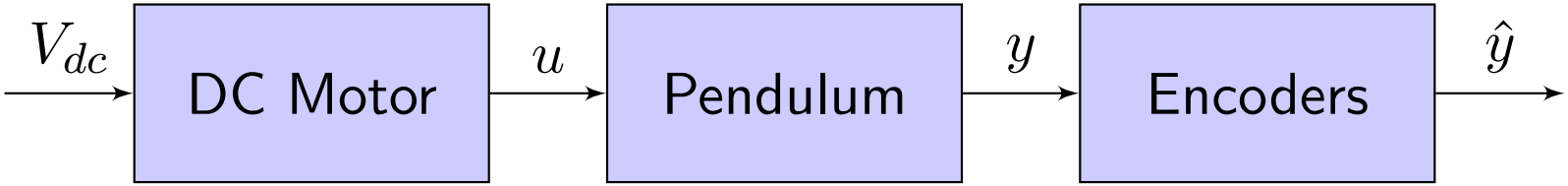
- rotary inverted pendulum
- ball and beam
- rotating masses
- magnetic levitation
- helicopter
- rotary flexible joint
- CNC machine
- linear inverted pendulum
- seesaw
- coupled water tanks

Rotary Inverted Pendulum

Rotary Inverted Pendulum

Rotary Inverted Pendulum

System



Quantities

- V_{dc} is the voltage applied to the motor
- u is the resulting torque applied to the horizontal arm
- y contains the angular position of the two arms
- $\hat{y}$ contains the *relative* angular position of the two arms

Main challenge

Dynamical model and instability of the upright equilibrium position of the pendulum

Control objectives

1. position control for the horizontal arm (FB)
2. stabilization of the vertical arm in the upright position (FB)
3. position control for the horizontal arm while keeping the vertical arm in the upright position (SS)
4. swing-up of the vertical arm (AC)

Ball and Beam

Ball and Beam

Ball and Beam

System

Quantities

- V_{dc} is the voltage applied to the motor
- u is the resulting torque applied to the gear of the lever arm
- y contains the gear angular position and ball position
- $\hat{y}$ contains the gear *relative* angular position and the ball *absolute* position on the beam

Ball and Beam

Main challenge

Marginal stability of equilibrium position of the ball on the beam and presence of state constraints

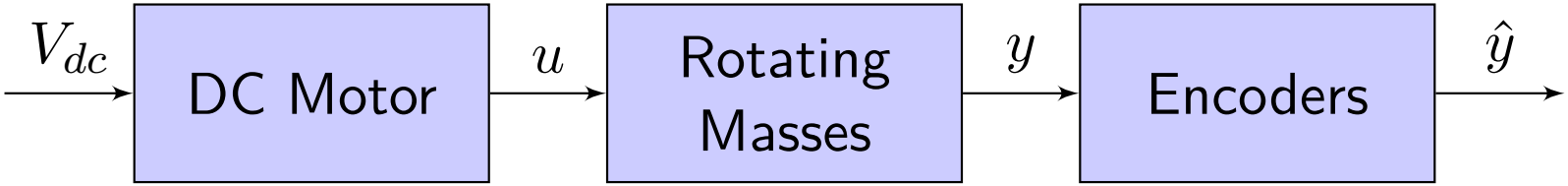
Control objectives

1. tilting control for the horizontal beam (FB)
2. ball position stabilization (FB & SS)
3. 1D ball trajectory tracking (SS & AC)

Rotating Masses

Rotating Masses

System



Quantities

- V_{dc} is the voltage applied to the motor
- u is the resulting torque applied to the gear shaft
- y contains the masses angular position
- $\hat{y}$ contains the masses *relative* angular position

Main challenge

Low damping of the system

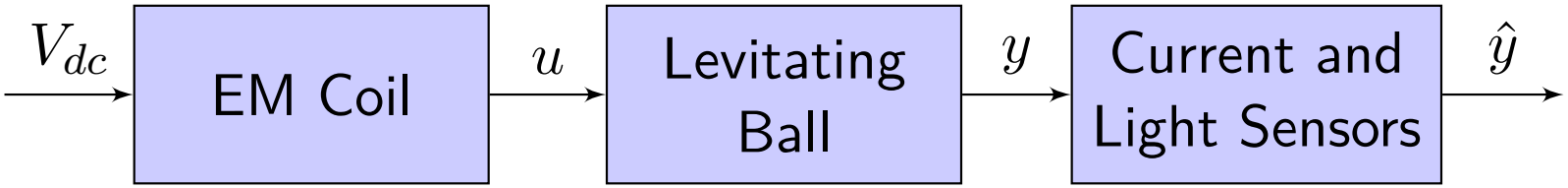
Control objectives

1. speed control for the first masses (FB)
2. position control for the first masses (FB & SS)
3. position control for the other two masses (SS & AC)

Magnetic Levitation

Magnetic Levitation

System



Quantities

- V_{dc} is the voltage applied to the electromagnet (EM)
- u is the resulting induced magnetic field
- y contains the coil current and the ball position
- $\hat{y}$ contains the measured current and *absolute* ball position

Main challenge

Highly nonlinear system and instability of the ball position

Control objectives

1. EM coil current control (FB)
2. ball position stabilization (FB & SS)
3. 1D ball trajectory tracking and lift-up (SS & AC)

Helicopter

Helicopter

Helicopter

System

Quantities

- V_{dc} is the voltage applied to the propellers' motors
- u is the resulting thrust generated by the propellers
- y contains the pitch and yaw angles
- $\hat{y}$ contains the pitch and yaw *relative* angles

Helicopter

Main challenge

Coupled MIMO nonlinear system

Control objectives

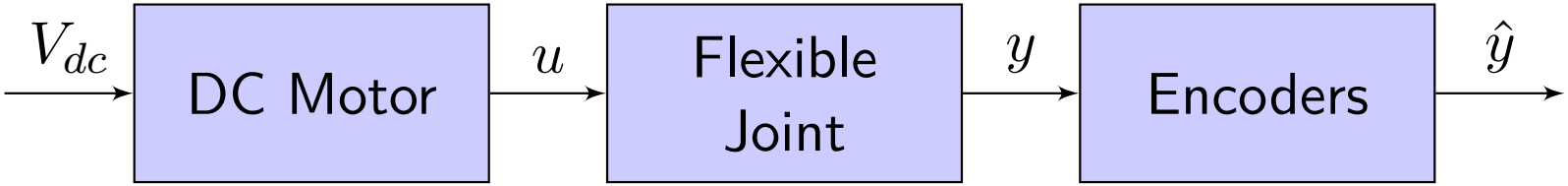
1. pitch control with yaw rotation locked (FB)
2. fixed attitude control (FB & SS)
3. reference attitude tracking (SS & AC)

Rotary Flexible Joint

Rotary Flexible Joint

Rotary Flexible Joint

System



Quantities

- V_{dc} is the voltage applied to the motor
- u is the resulting torque applied to the top base
- y contains top base and arm angular positions
- $\hat{y}$ contains the top base and arm *relative* angular position

Main challenge

Low damping of the system and variable parameters

Control objectives

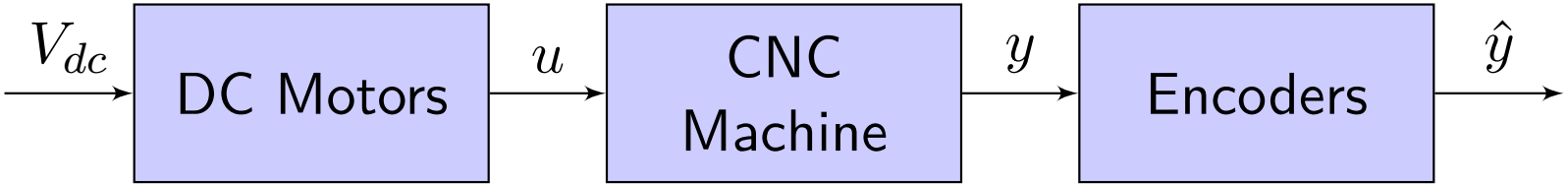
1. position control for the top base (FB)
2. position control for the the arm tip (FB & SS)
3. position control for the the arm tip with uncertainty in the spring stiffness and arm moment of inertia (SS & AC)

CNC Machine

CNC Machine

CNC Machine

System



Quantities

- V_{dc} contains voltages applied to both motors
- u contains the resulting torques acting on the chain
- y contains the coordinates of the end effector
- $\hat{y}$ contains the motors' *relative* angular position

CNC Machine

Main challenge

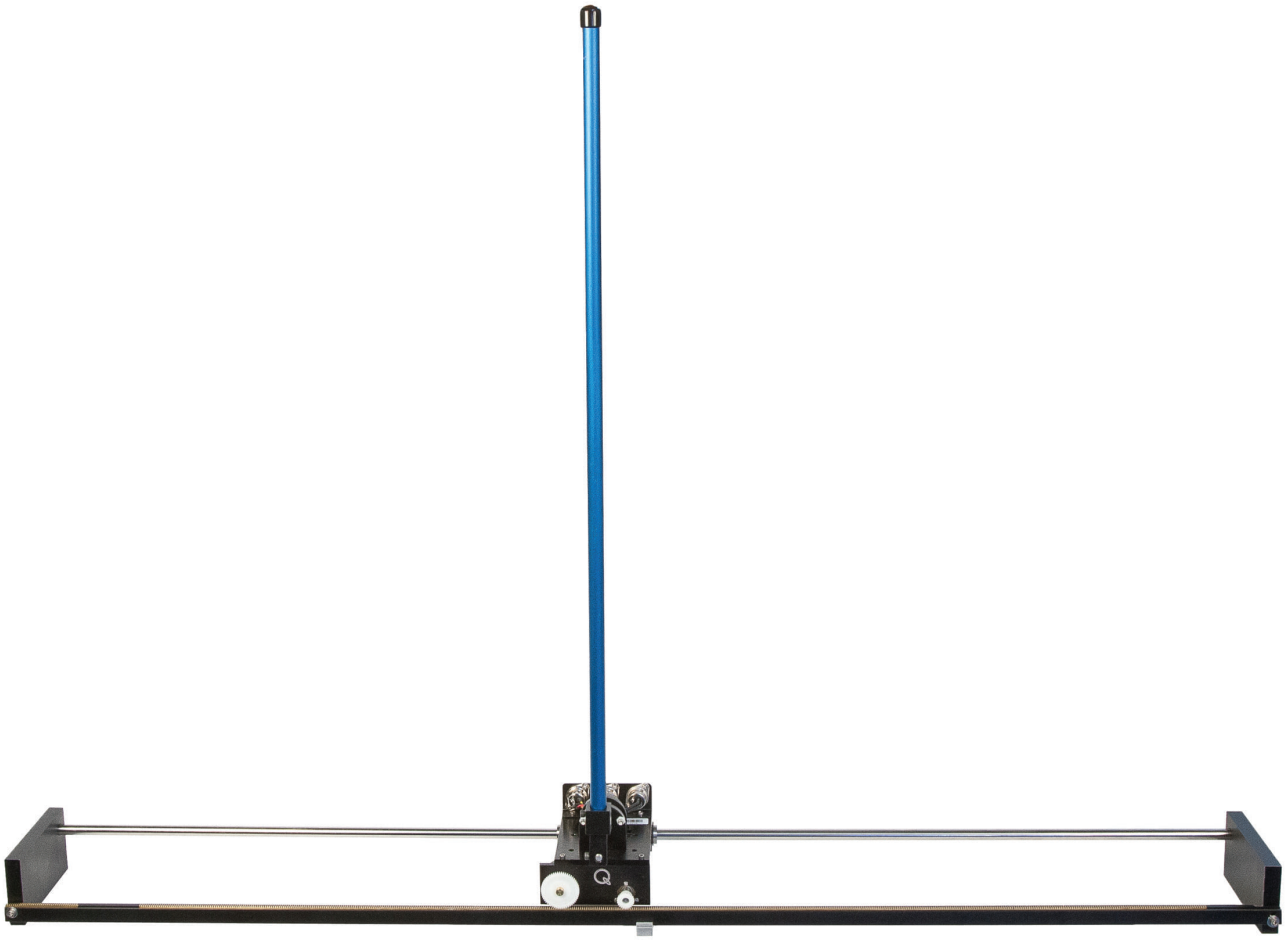
Coupled MIMO nonlinear system

Control objectives

1. motor position control (FB)
2. end effector 2D position control (FB & SS)
3. 2D trajectory tracking (SS & AC)

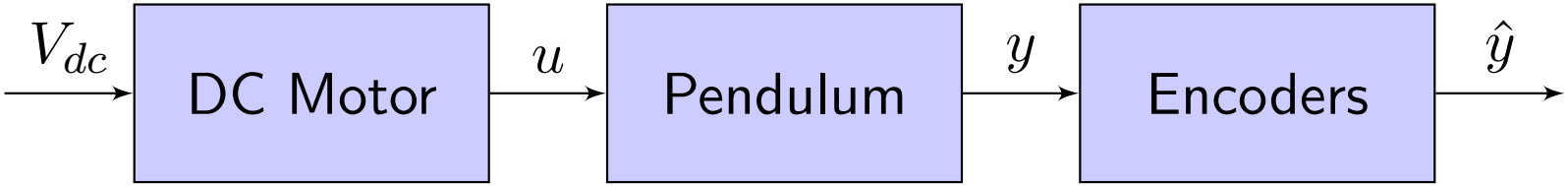
Linear Inverted Pendulum

Linear Inverted Pendulum



Linear Inverted Pendulum

System



Quantities

- V_{dc} is the voltage applied to the motor
- u is the resulting force applied to the cart
- y contains the cart position and the arm angular position
- $\hat{y}$ contains the *relative* cart and arm positions

Linear Inverted Pendulum

Main challenge

Dynamical model and instability of the upright equilibrium position of the pendulum

Control objectives

1. position control for the cart (FB)
2. stabilization of the arm in the upright position (FB)
3. position control for the cart while keeping the arm in the upright position (SS)
4. swing-up of the arm (AC)

Seesaw

Seesaw

Seesaw

System

Quantities

- V_{dc} is the voltage applied to the motor
- u is the resulting force applied to the cart
- y contains the cart position and the body angular position
- $\hat{y}$ contains the *relative* cart and body positions

Seesaw

Main challenge

Dynamical model and instability of the horizontal equilibrium position of the body

Control objectives

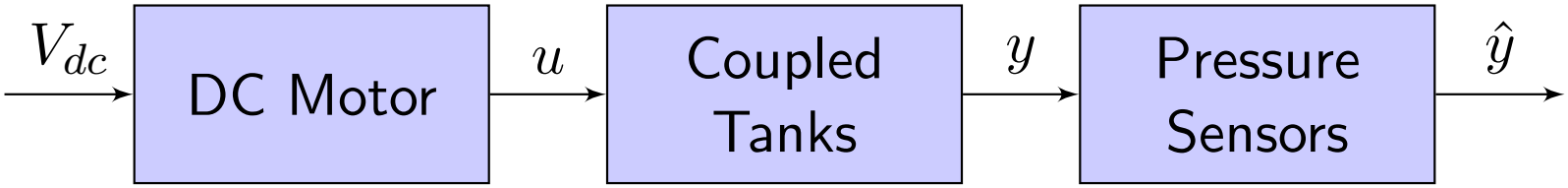
1. position control for the cart (FB)
2. stabilization of the body in the horizontal position (FB & SS)
3. periodic trajectory tracking for the body and lift-up (SS & AC)

Coupled Water Tanks

Coupled Water Tanks

Coupled Water Tanks

System



Quantities

- V_{dc} is the voltage applied to the pump
- u is the resulting flow rate at the tank inlet
- y contains the water level into the two tanks
- $\hat{y}$ contains the measured pressure at the bottom of the two tanks

Main challenge

Fairly nonlinear system with state and input constraints and variable parameters

Control objectives

1. level control for tank 1 (FB)
2. level control for tank 2 (in series with tank 1) (FB & SS)
3. level control for tank 2 (in series with tank 1) with uncertainty in the tanks outlet flows (SS & AC)

Laboratory environment

Laboratory environment

MATLAB + SIMULINK

Note

We will use R2024b: be careful to avoid compatibility issues

Laboratory environment

PC $\Longleftrightarrow$

$\Longleftrightarrow$ $\Updownarrow$

Set-up $\Longleftrightarrow$

Material provided

For each set-up there is a zip file (`NameOfTheSetup.zip`) on WeBeep containing the following files:

- datasheet for the data acquisition board (DAQ)
- datasheet for the power amplifier (PA)
- datasheet for the DC motor and the gearbox
- datasheet for the specific set-up
- simulink template file containing appropriate input/output blocks to start experimenting right away

Note

Some experiments connect to the PC directly via USB and the zip file contains one datasheet only and the simulink template

Quick-start guide

Example

In the following assume a group received the *flexible joint*

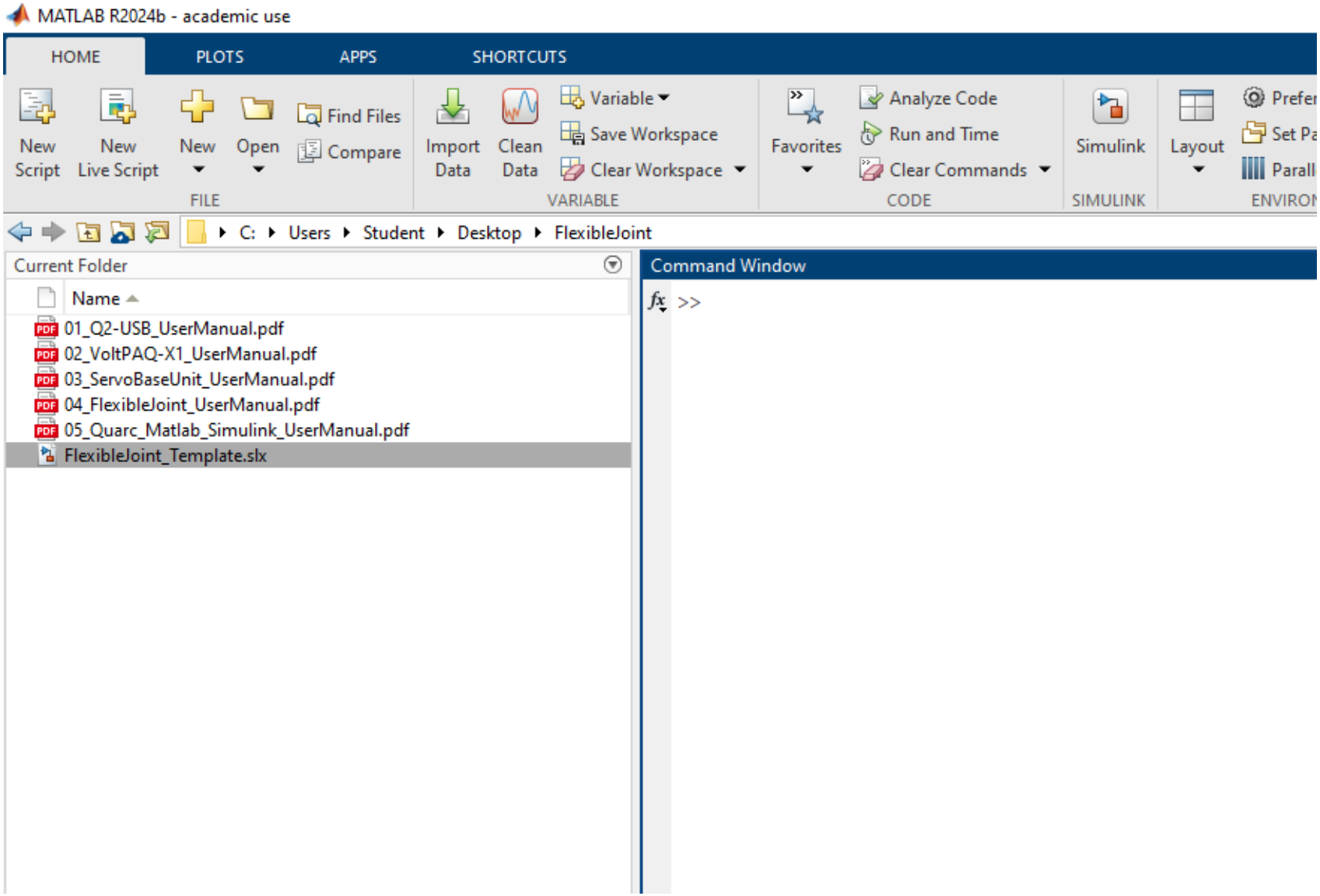
Beginning of the lecture

- turn on the lab PC (use the "Student" account)
- connect the DAQ board to a USB3 socket
- connect PA to the nearest AC plug and turn it on

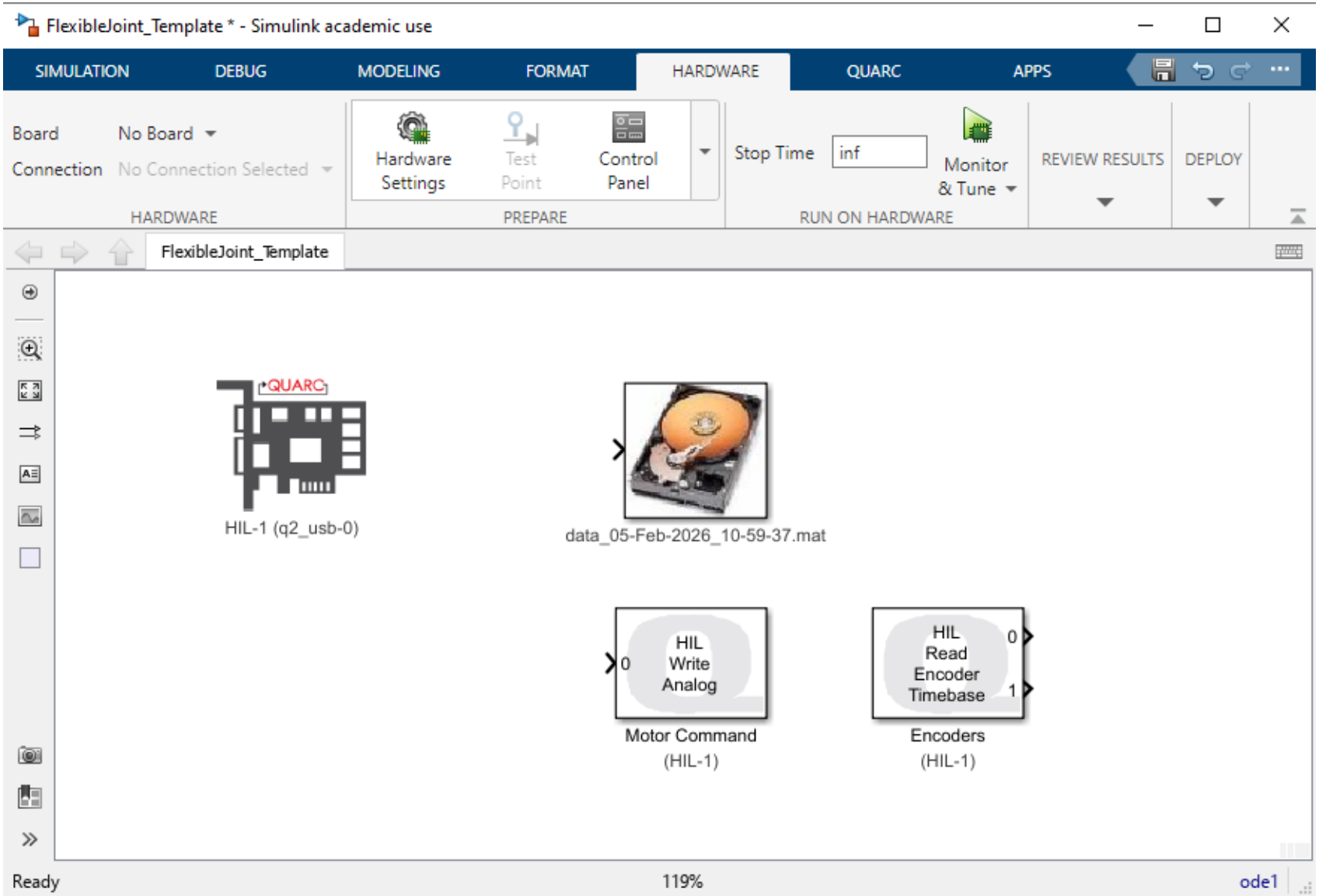
First steps for using the set-up

- open the browser and go to the WeBeep portal
- login with your credentials, download `FlexibleJoint.zip`
- drag the folder inside the zip and drop it onto the desktop
- open MATLAB R2024b
- click on the icon in the top-left part of the screen and navigate to the `FlexibleJoint` folder

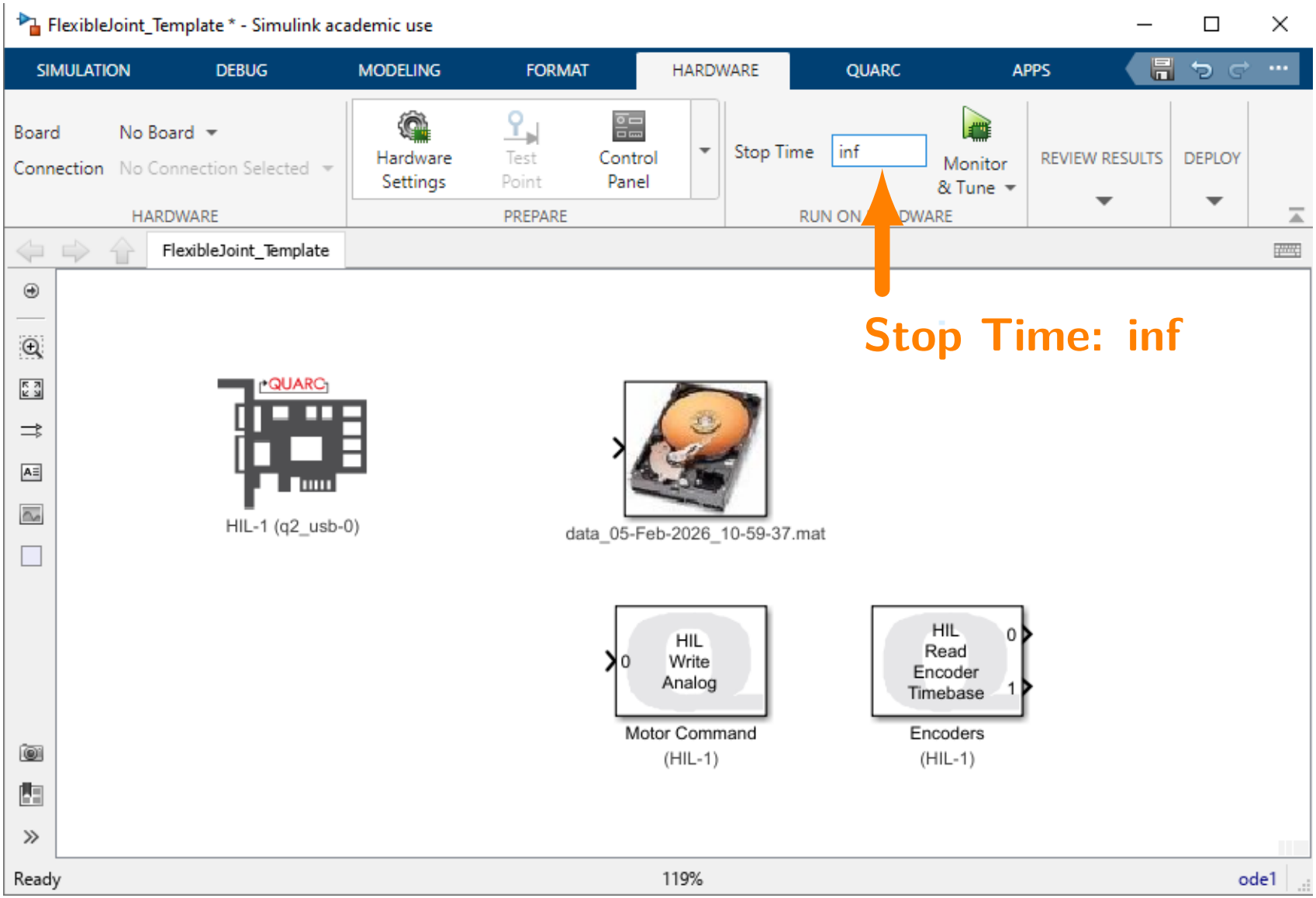
Quick-start guide



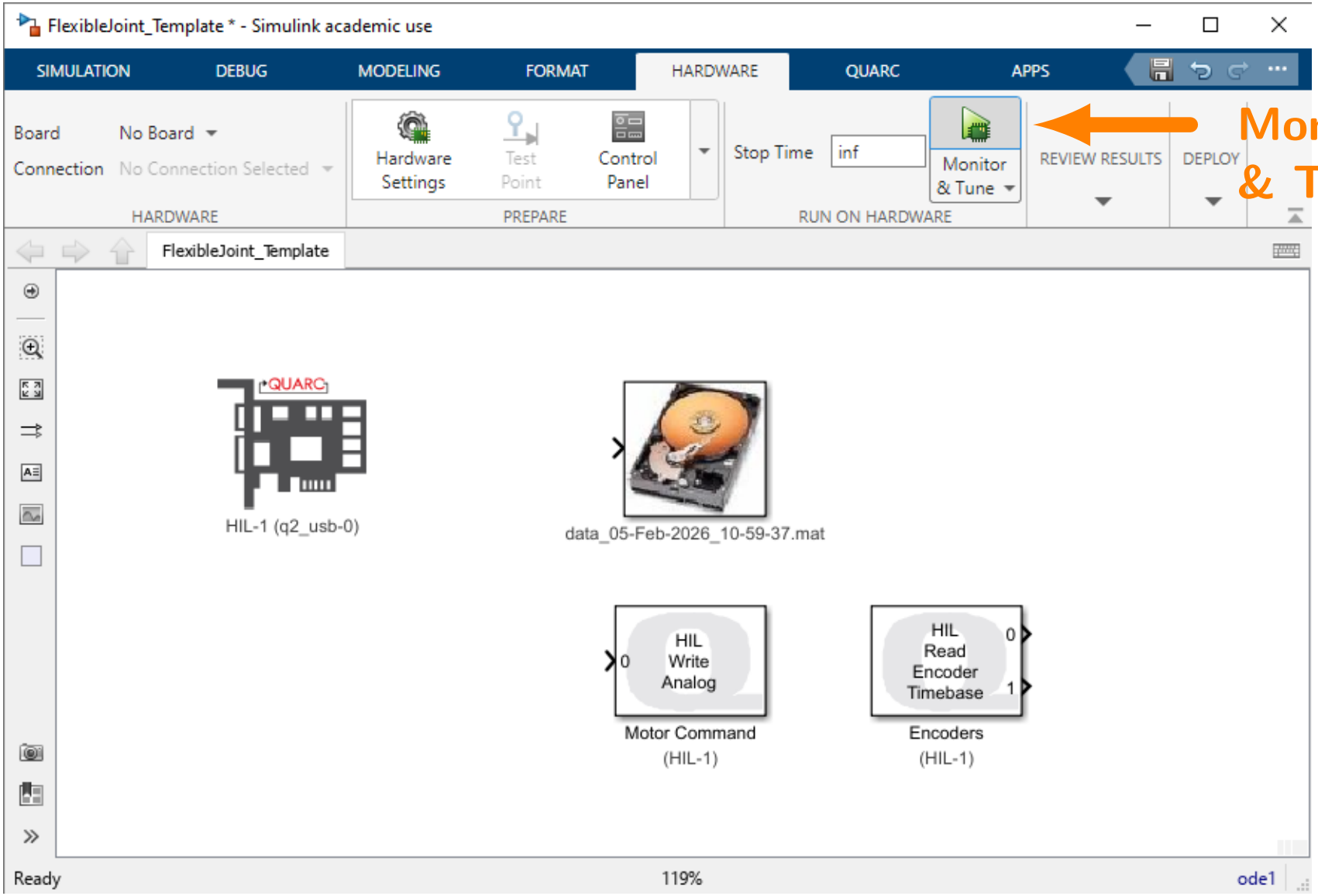
Quick-start guide



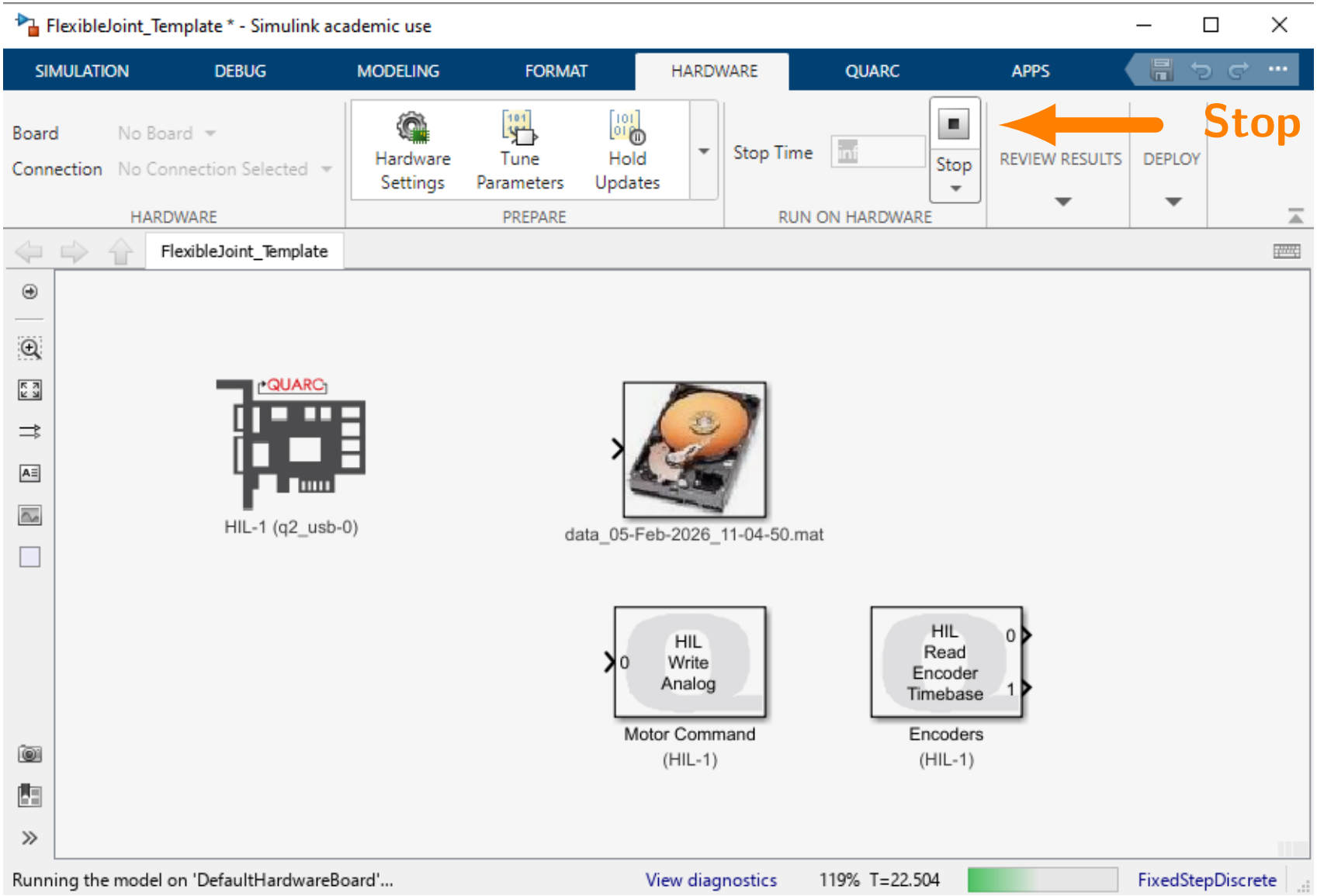
Quick-start guide



Quick-start guide



Quick-start guide



Note

While the Simulink model is running you can change constant values, amplitude/frequency of some signals, values of gains, etc.

Other blocks

There are other blocks in the QUARC library and you can refer to their documentation for additional information
https://docs.quanser.com/quarc/documentation/index.html

Asking for help

- if you need quick technical assistance call the tutor first
- if you have a conceptual doubt or you are lost call the teacher

Warnings

- the first time, ask the teacher before running the Simulink model on the real set-up
- keep away from all moving parts (gears, arms, etc.) when experiment is running
- remember that set-ups are real, please be careful when you are performing tests to avoid any damage

End of the lecture

- make sure you stop all running Simulink models
- close Matlab
- zip all of your files
- transfer them to a personal device (thumb-drive or cloud)
- delete all files on the lab PC
- shut down the lab PC
- switch off the PA and remove the AC plug
- disconnect the DAQ from the PC
- collect all cables under the side table housing the set-up

Using AI

Policy

The use of AI is not forbidden, but it requires caution

LLMs are text generators, not truth generators

The narrower the topic, the less training data they have, and the more likely they are to produce imprecise answers

Finding reasoning errors is harder than thinking yourself

You should do the thinking and only use the AI for small checks if you are unsure, not the other way around

Enjoy the course! =)